

ECSE-4040

Digital Electronics

Lecture 1

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Lecture Outline

- Course Mechanics and Logistics
- Digital IC Evolution (ITRS Roadmap)
- Combinatorial and Sequential Logic Gates
- DSM Physical Design Flow
- MOSFET Evolution

DEI Course Syllabus

(Modified) Catalog Description

Analysis and design of switching-mode circuits: NMOS, CMOS, RTL, DTL, TTL, and ECL digital-logic families. Topics include: basic logic gates (voltage transfer characteristics, noise margin, fan in-fan out, propagation delay, power dissipation), flip flops, Schmitt triggers, oscillators, timers, memories, A/D and D/A converters, and optional advanced topics.

Objectives

Study in detail the principles of operation and, in particular, the electrical characteristics, as well as various models of basic semiconductor devices (mainly, diodes and MOSFET's).

Prerequisites

Basic knowledge of circuits and digital logic, Introduction to Electronics (ECSE-2050) and Computer Components and Operation (ECSE-2610) or equivalent or permission of the instructor.

DEI Course Syllabus

Textbook

Donald A. Neamen, *Microelectronics: Circuit Analysis and Design*, 4th Ed., McGraw-Hill, 2010.

Reference Books

D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.

N.H.E. Weste and D.M. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, Addison-Wesley, 2011.

J.M. Rabaey, A. Chandrakasan, and B. Nikolic, *Digital Integrated Circuits: A Design Perspective*, 2nd Ed., Prentice-Hall, 2003.

Method of Instruction

The course will consist of a series of lectures and a weekly simulation laboratory based on SPICE. The lectures will cover the content of the text and other references scheduled in the syllabus. The student is expected to have read the material prior to class and to take part in discussing the material during lecture.

DEI Course Syllabus

Computing

LT SPICE will be used for circuit simulations. The students are expected to be already familiar with it from his/her previous lower-level undergraduate courses.

Format

Classes.....15 weeks
Simulation Lab.....12 weeks
Lectures.....29 sessions

Grading

Studio Problems.....30%
Mid-Term Quiz.....30%
Final Exam.....40%

The grading of problems/quiz will be assumed final a week after they are handed back.

DEI Course Calendar

Text: D.A. Neamen, *Microelectronics: Circuit Analysis and Design*, 4th Ed., McGraw-Hill, 2010.

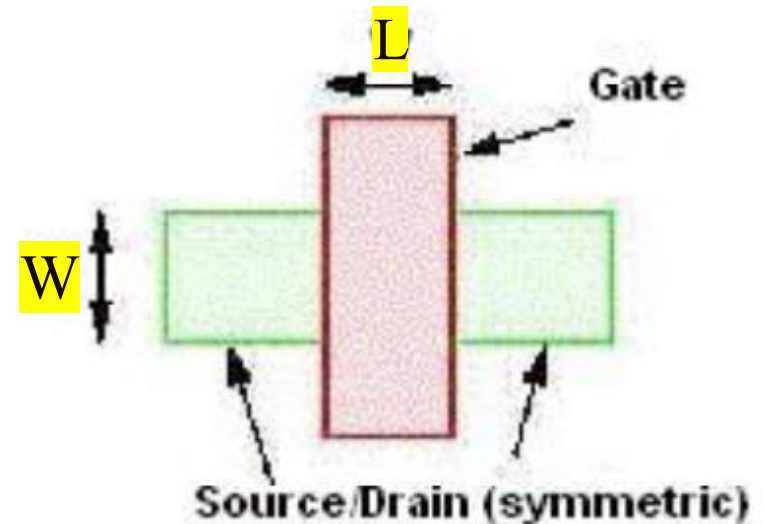
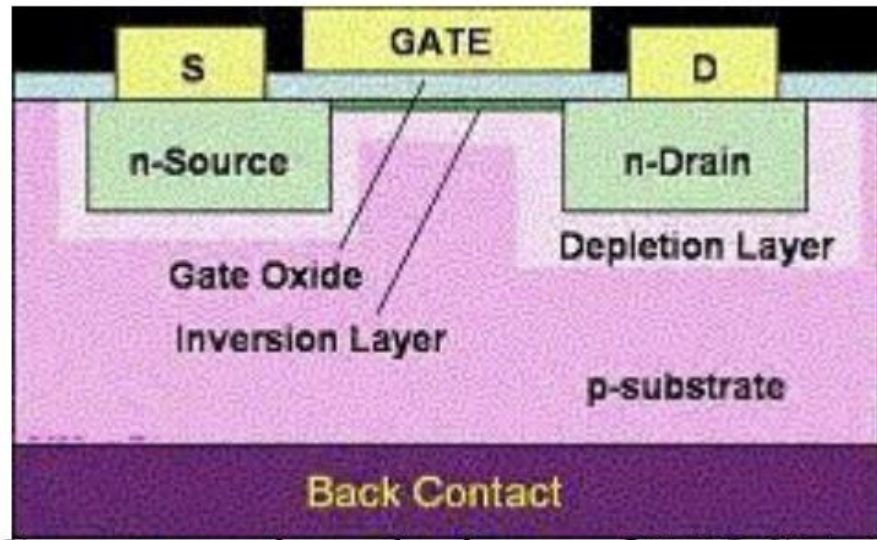
Wk	Date	Topic	Readings	Assignments
Wk 1	1/6	Lecture: Tu, Introduction; F, MOSFETs.	Neamen: pp. 1141 – 1144; LTspice Web Site (www.linear.com/ltspice).	Studio: W, No Class.
Wk 2	1/13	Lecture: Tu, MOSFETs; F, MOSFETs.	Neamen: Ch 3. pp. 125 – 146; HJS 3 rd Ed: Ch. 2 pp. 35 – 64.	Studio: W, LT SPICE Tutorial; P2.1, 2.5 (HJS 3 rd Ed.)
Wk 3	1/20	Lecture: Tu, MOSFETs (<i>Parameters, Models</i>); F, MOS Gates (<i>VTC, Noise Margins</i>).	Neamen: Ch. 4 pp. 205 – 215; HJS 3 rd Ed: Ch. 3 pp. 111 – 138.	Studio: W, Parameter extraction, P3.5 (HJS 3 rd Ed.).
Wk 4	1/27	Lecture: Tu, MOS Gates (<i>Inverters</i>); F, MOS Gates (<i>Sizing, VTC, NAND, NOR</i>).	Neamen: Ch. 16 pp. 1147 – 1202; HJS 3 rd Ed: Ch. 4 pp 142 – 180, Ch. 5 pp. 195 – 205.	Studio: W, P4.1 & 4.2 (HJS 3 rd Ed.).
Wk 5	2/3	Lecture: Tu, MOS Gates (<i>RC Delays, Combinational Logic</i>); F, MOS Gates (<i>Noise Sources, NM-Delay, Sequential Logic</i>).	HJS 3 rd Ed: Ch. 4 pp 181 – 186, Ch. 5 pp. 206 – 227.	Studio: W, P4.10 & 4.13 (HJS 3 rd Ed.). <u>MID-TERM EXAM STARTS</u> (at the end of Studio).
Wk 6	2/10	Lecture: Tu, MOS Gates (<i>Sequential Logic, Power</i>); F, MOS Gates (<i>Power</i>)	HJS 3 rd Ed: Ch. 5 pp. 228 – 242.	Studio: W, P4.14, 5.1, & 5.7 (HJS 3 rd Ed.).
Wk 7	2/17	Lecture: Tu, No Class ; F, MOS Gates (<i>Propagation Delay, Capacitance</i>)	HJS 3 rd Ed: Ch. 2 pp. 79 – 83, Ch. 6 pp. 249 – 267.	Studio: W, P5.12 & 5.16 (HJS 3 rd Ed.).
Wk 8	2/24	Lecture: Tu, MOS Gates (<i>Input Slope, Elmore Delay</i>); F, Interconnects.	HJS 3 rd Ed: Ch. 6 pp. 267 – 277; Ch. 3 pp. 100 – 103, Ch. 10 pp. 441 – 468.	Studio: W, P6.4 (HJS 3 rd Ed) <u>MID-TERM EXAM DUE</u>
Wk 9	3/10	Lecture: Tu, BJT (<i>Ebers-Moll Model</i>); F, BJT Inverter (<i>VTC</i>).	Neamen: Ch. 5 pp. 285 – 326; HJ 2 nd Ed: Ch. 5 pp. 108 – 124, Ch. 6 pp. 126 – 147.	Studio: W, P5.1, 5.2, & P6.1 – 6.4 (HJ 2 nd Ed.).
Wk 10	3/17	Lecture: Tu, BJT Gates (<i>ECL</i>); BJT Gates (<i>Low Power ECL</i>).	Neamen: Ch. 17 pp. 1256 – 1276; HJ 2 nd Ed: Ch. 7 pp. 173 – 179, Ch. 8 pp. 192 – 194.	Studio: W, P7.9– 7.12 (HJ 2 nd Ed.).
Wk 11	3/24	Lecture: Tu, BJT Gates (<i>CML</i>); Regenerative Ckts (<i>Mono & Astable</i>).	HJ 2 nd Ed: Ch. 8 pp. 202 – 209.	Studio: W, P8.1 & 8.2 (HJ 2 nd Ed).
Wk 12	3/31	Lecture: Tu, Memories (<i>SRAM</i>); F, Memories (<i>DRAM, Sense Amplifiers</i>)	Neamen: Ch. 16 pp. 1208 – 1221; HJS 3 rd Ed: Ch. 8 pp. 368 – 377; WH 4 th Ed: Ch. 12 pp. 498 – 526	Studio: W, P7.5--7.8.
Wk 13	4/7	Lecture: Tu, Memories (<i>TG Logic</i>); F, Memories (<i>SDRAM, Peripherals</i>)	HJS 3 rd Ed: Ch. 7 pp. 309 – 333, Ch. 8 pp. 359 – 368, 377 – 394	Studio: W
Wk 14	4/14	Lecture: Tu, F (<i>D/A & A/D Converters</i>)	Neamen: Ch. 16 pp. 1226 – 1235	Studio: W, No Class
Wk 15	4/21	Lecture: Tu, Review, Th, F, Study Days		Studio: W, In-Class Final FINAL EXAM DUE

Circuit Design Priorities

- Requirements
- Functionality
- Timing
- Power
- Circuit Noise Tolerance
- Chip Area
- Cost
- Time-to-Market
- Supply Variations
- Testability
- Packaging
- Process Variations
- Yield
- Temperature Variations
- Short/Long Term Reliability

MOSFET

Metal-Oxide-Semiconductor Field-Effect Transistor



MOS IC History

MOSFET scaling (process nodes)

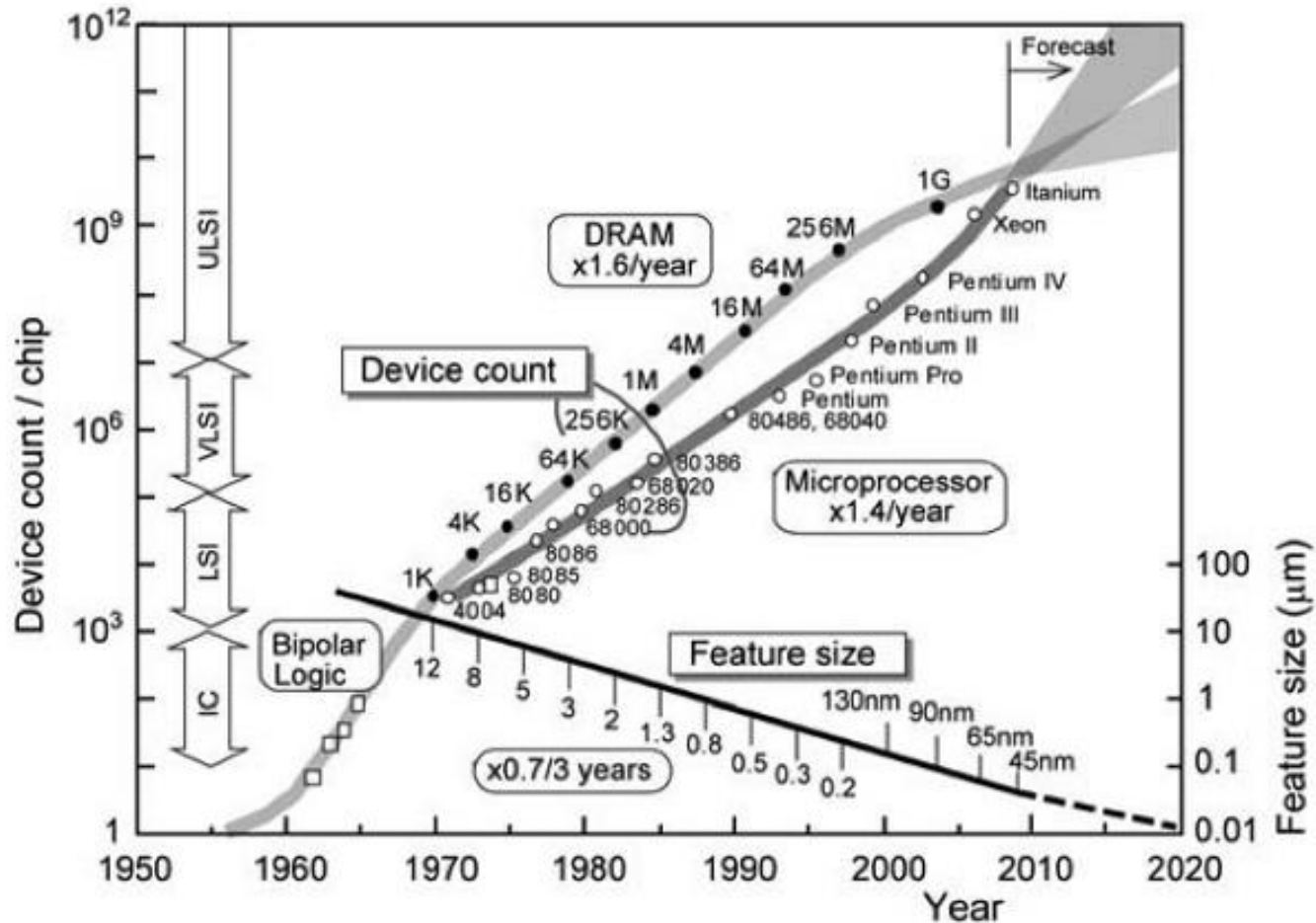
10 μm – 1971
6 μm – 1974
3 μm – 1977
1.5 μm – 1981
1 μm – 1984
800 nm – 1987
600 nm – 1990
350 nm – 1993
250 nm – 1996
180 nm – 1999
130 nm – 2001

90 nm – 2003
65 nm – 2005
45 nm – 2007
32 nm – 2009
22 nm – 2012
14 nm – 2014
10 nm – 2016
7 nm – 2018
5 nm – 2020

Future

3 nm – ~2022
2 nm – >2023

Technology Evolution



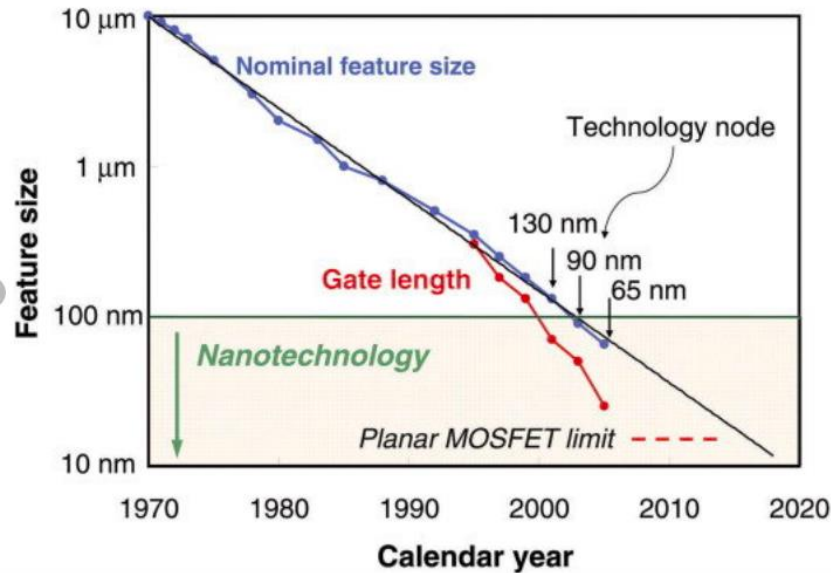
ITRS Roadmap

Table 1.2

Technology roadmap based on the ITRS

Year	Technology Node (nm)	V_{DD} (V)	V_{th} (V)	t_{ox} ($\text{\AA} = 10^{-10} \text{ m}$)	No. of Transistors
1995	350	3.3	0.6	75	10M
1997	250	2.5	0.55	50	20M
1999	180	1.8	0.5	35	40M
2001	130	1.2	0.4	22	100M
2004	90	1.0	0.35	20	250M
2007	65	0.7	0.3	14	500M
2010	45	0.6	0.25	11	1B
2013	32	0.5	0.22	10	2B
2016	22	0.4	0.2	9	4B

Technology Node vs. Gate Length

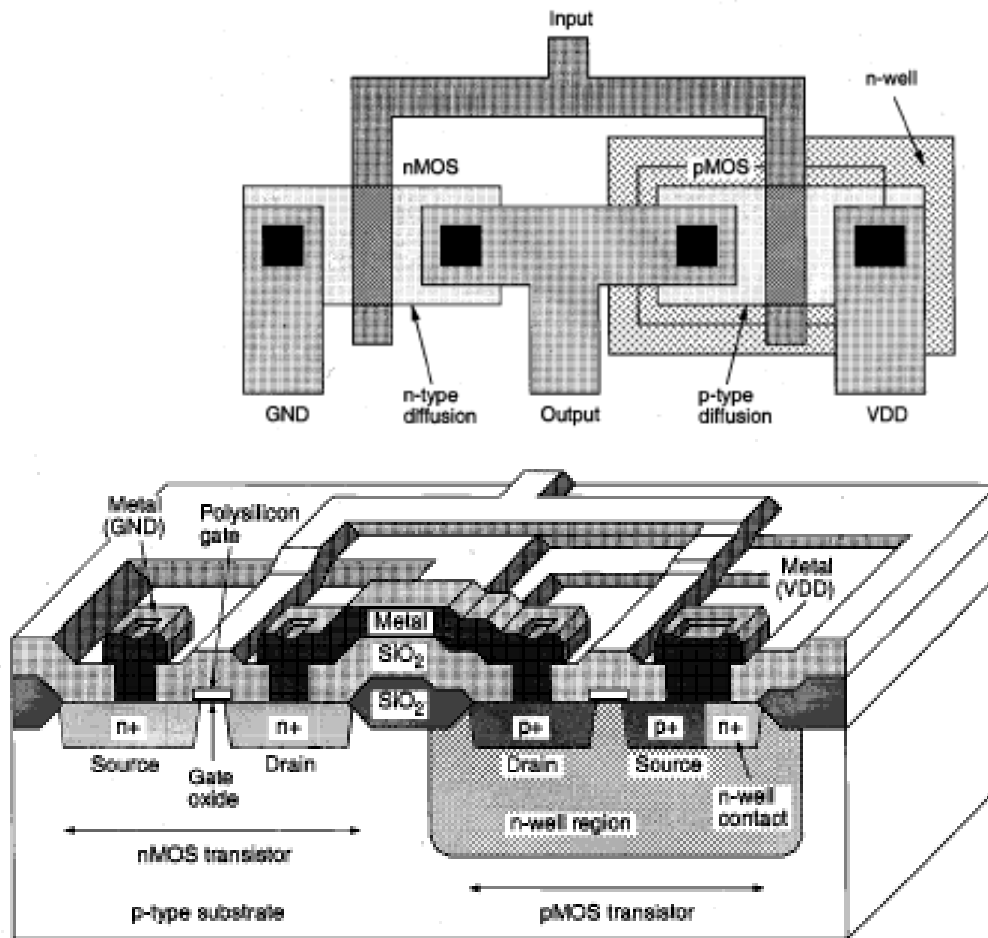


<https://www.hpcwire.com/2016/07/28/transistors-wont-shrink-beyond-2021-says-final-its-report/>

YEAR OF PRODUCTION	2015	2017	2019	2021	2024	2027	2030
Logic device technology naming	P70M56	P48M36	P42M24	P32M20	P24M12G1	P24M12G2	P24M12G3
Logic industry "Node Flange" Labeling (nm)	"16/14"	"11/10"	"8/7"	"6/5"	"4/3"	"3/2.5"	"2/1.5"
Logic device structure options	finFET FDSOI	finFET FDSOI	finFET LGAA	finFET LGAA VGAA	VGAA, M3D	VGAA, M3D	VGAA, M3D
	 	 	 	 	 	 	 

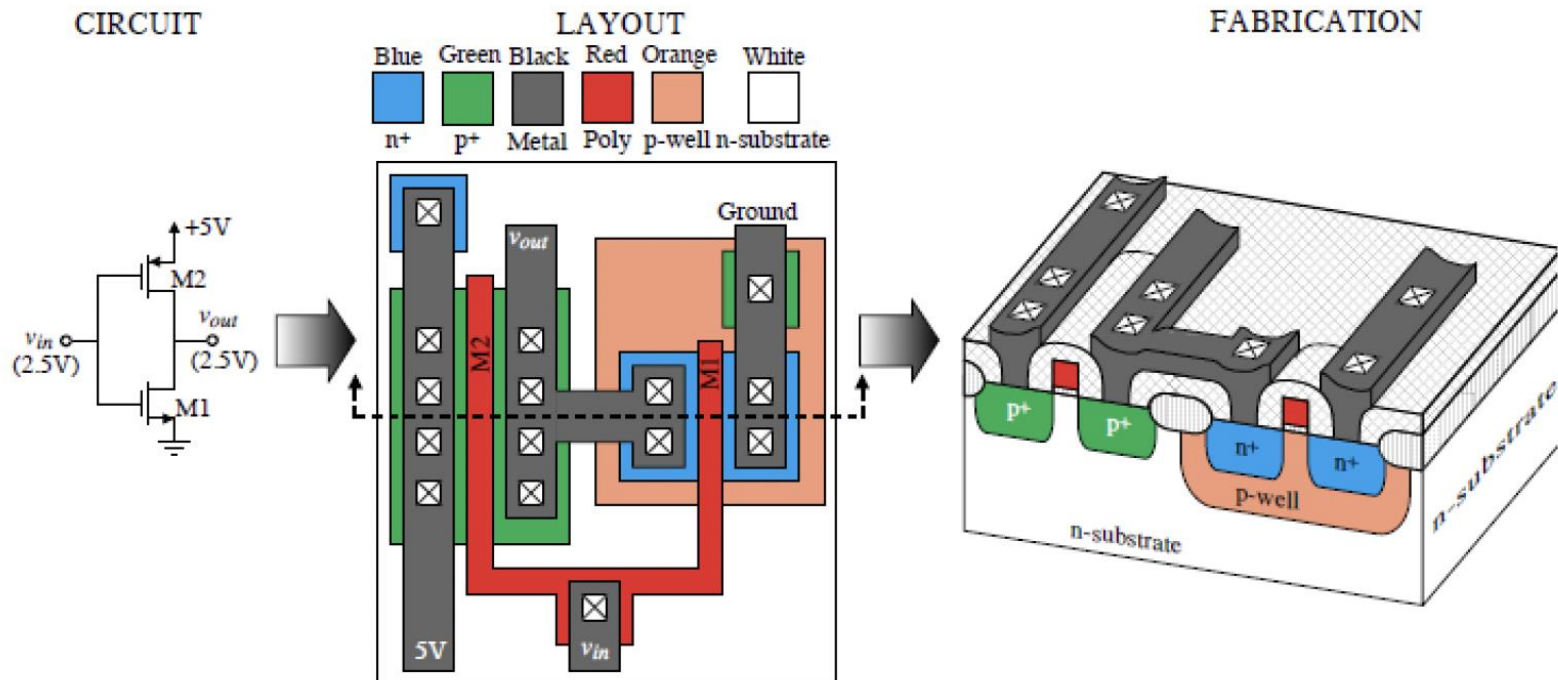
Figure MM3: Transistor structure roadmap: FDSOI, finFET, lateral nanowire, vertical nanowire, and monolithic 3D.

Gate Length or Gate Width?



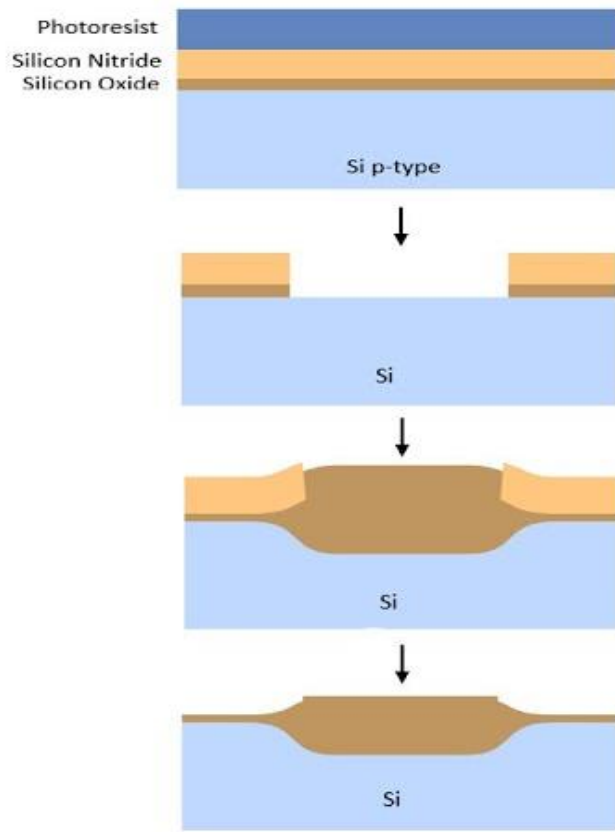
Gate Length or Gate Width?

- Physical design is the process of representing the electrical design in **a layout** consisting of many distinct geometrical rectangles at various levels.
- The layout is then used to create the actual, three-dimensional integrated circuit through a process called **fabrication**.

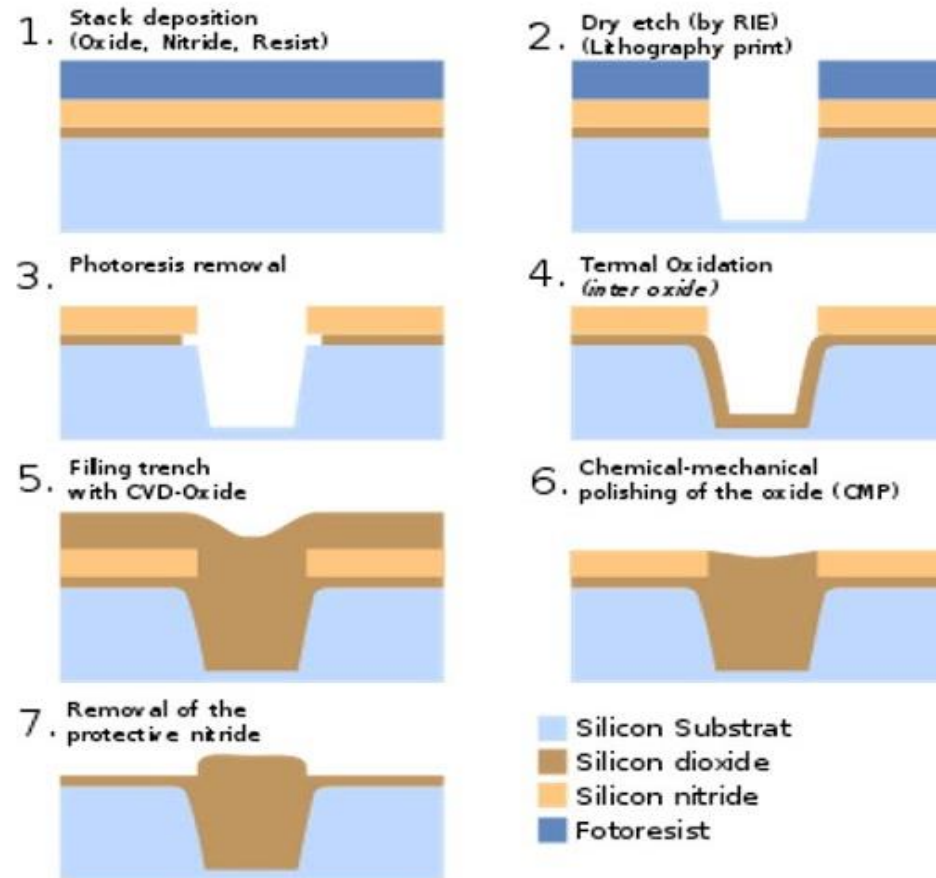


Device Isolation Techniques

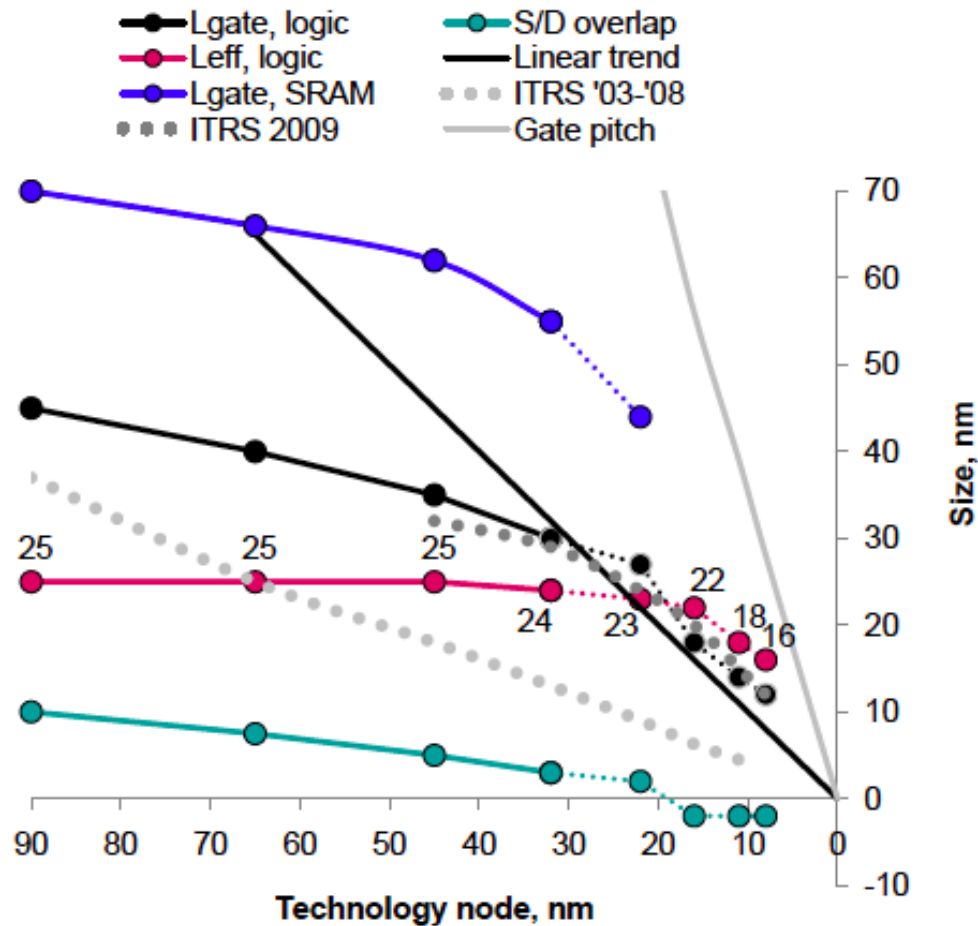
Local Oxidation of Silicon



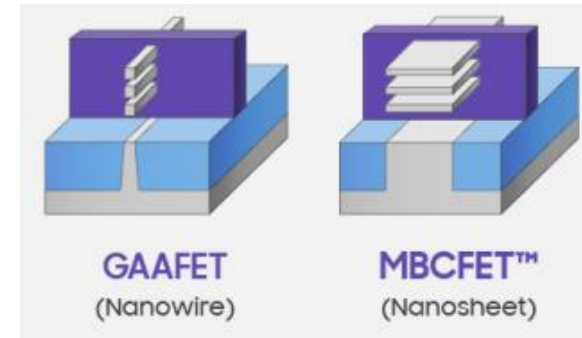
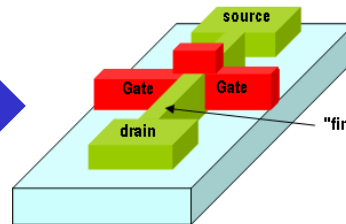
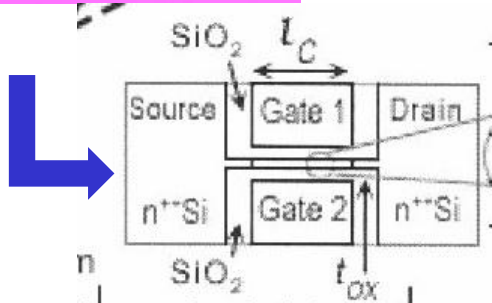
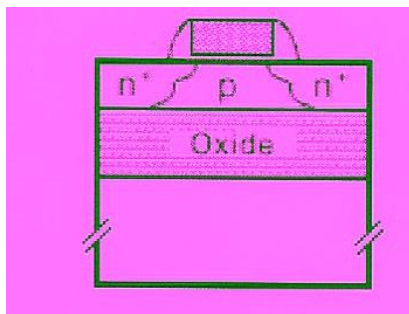
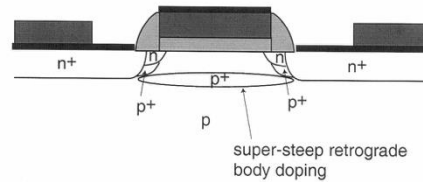
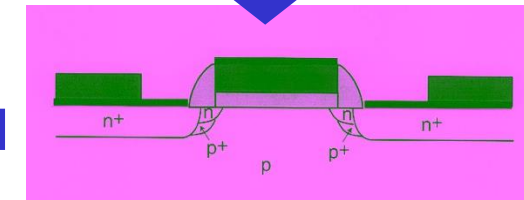
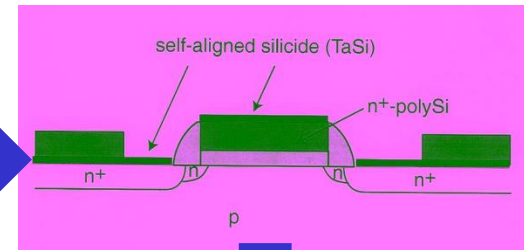
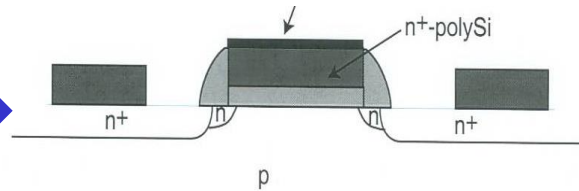
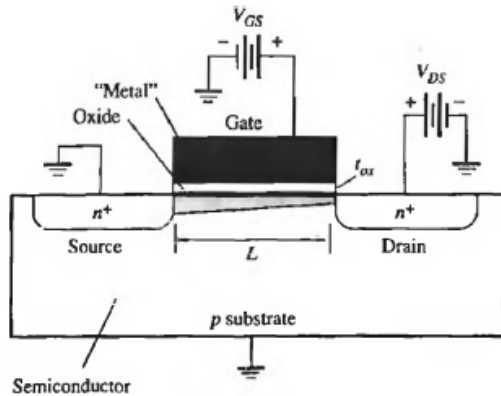
Shallow Trench Isolation



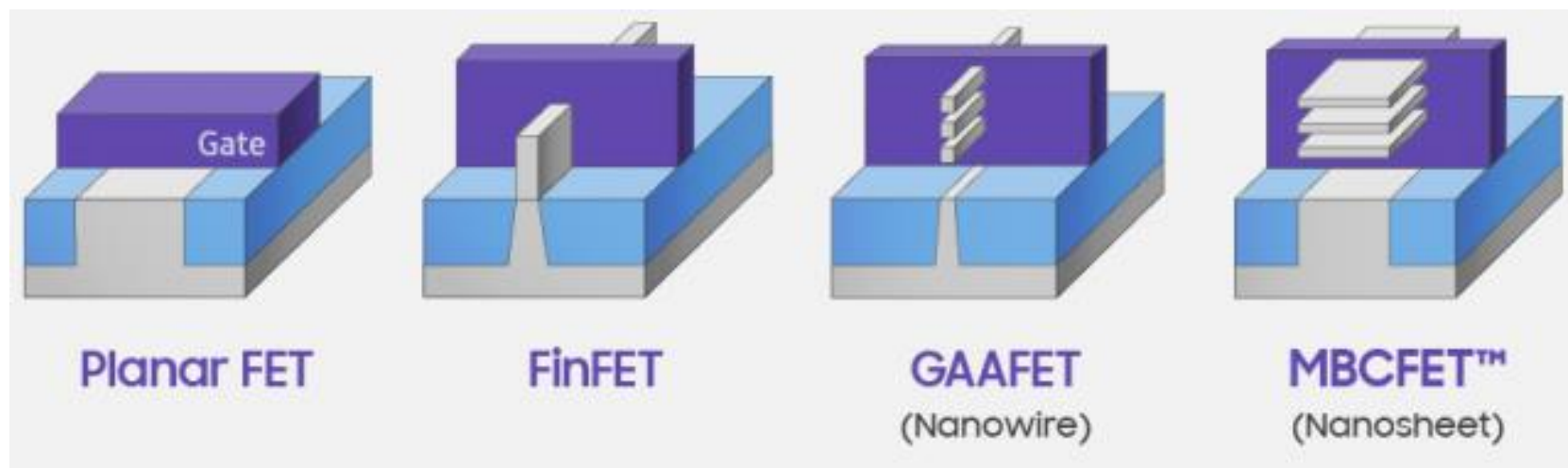
MOSFET Evolution



MOSFET Evolution



MOSFET Evolution



DSM Device and Interconnect Issues

Table 1.1

A short list of major DSM device and interconnect issues

DSM Devices

Short-channel effects on V_T

Velocity saturation

Thin-oxide (tunneling/breakdown)

Subthreshold current

DIBL

Hot-carrier effects

DSM Wires

RC delays

IR drop

Ldi/dt

Capacitive coupling

Inductive coupling

Electromigration

Antenna effects

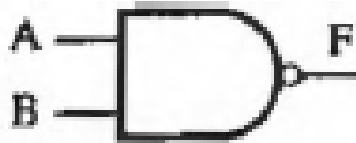
Basic Logic Gates

Inverter (INV)



A	F
0	1
1	0

NAND



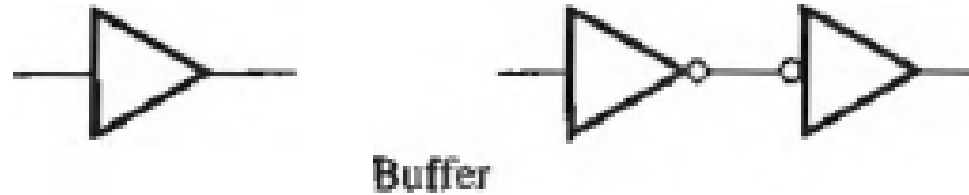
A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

NOR



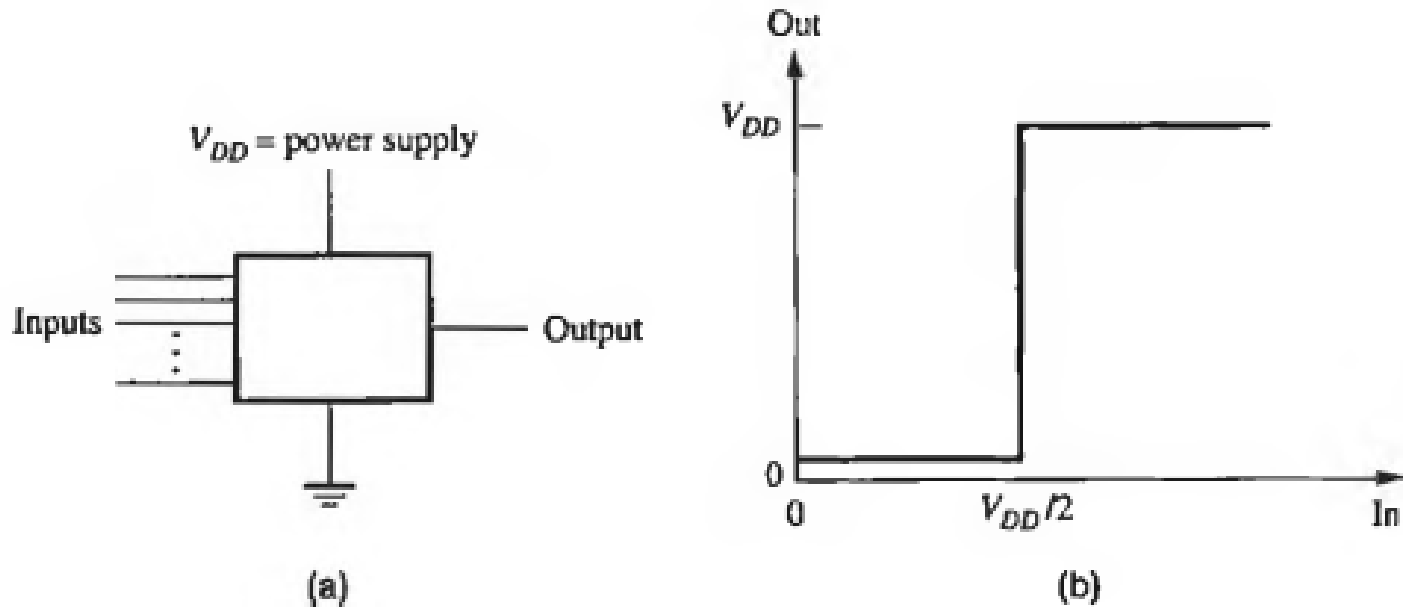
A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

Alternative Rep. of Buffer, NOR, NAND Gates



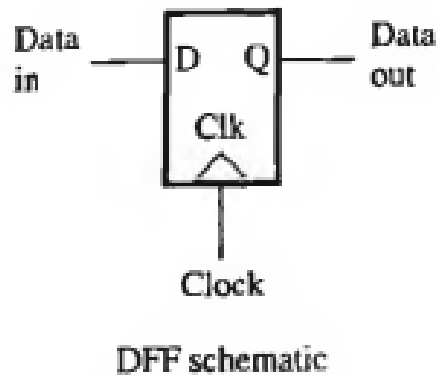
Constructed using DeMorgan's Law

Ideal Digital Circuit



- Single Power Supply (Minimum Power Consumption)
- Low Output Impedance; High Input Impedance
- Abrupt Transition Voltage at $V_{DD}/2$
- Negligible Time Delay

Positive-edge Triggered D-Type Flip Flop



Clk	D	Q (old)	Q (new)
0	X	0	0
0	X	1	1
0 → 1	0	X	0
0 → 1	1	X	1
1	X	0	0
1	X	1	1
1 → 0	X	0	0
1 → 0	X	1	1

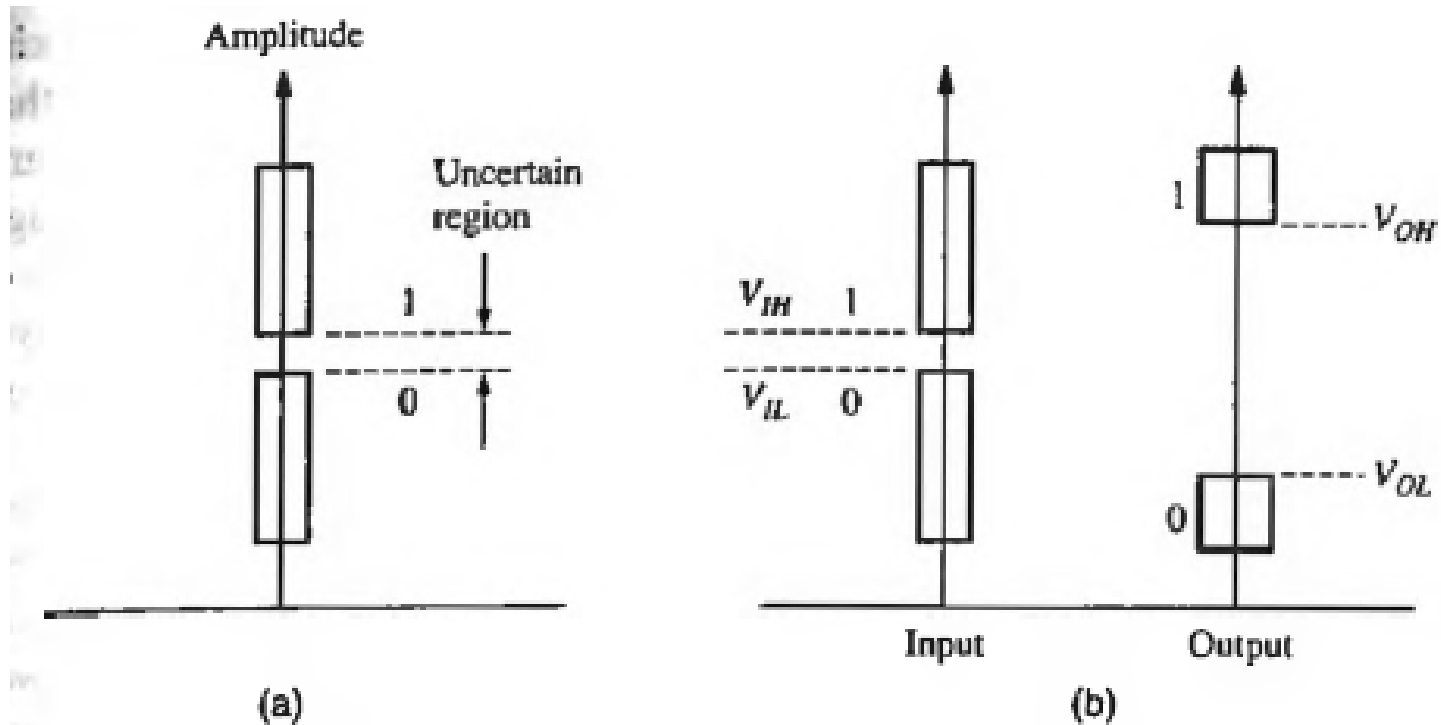
Excitation table

X – “Don’t Care”

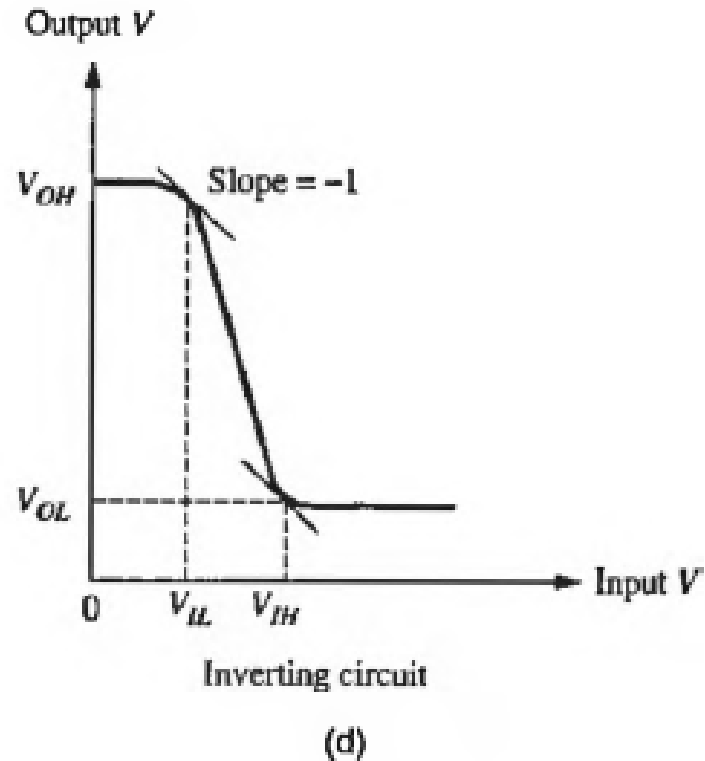
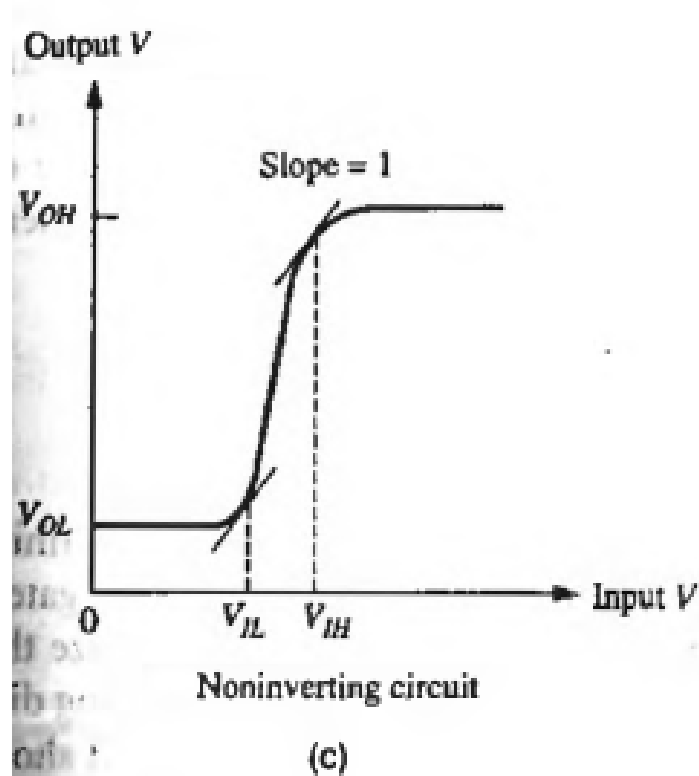
Properties of Digital Circuits

- Deterministic: Binary Input(s) give Binary Output
- Quantized Amplitude
- Amplitude Regeneration
- Directivity: Unilateral Operation (Changing Output cannot be from Unchanging Input(s))
- Multiple Fan-ins and Fan-outs

Logic Abstraction



Voltage Transfer Curve (VTC)

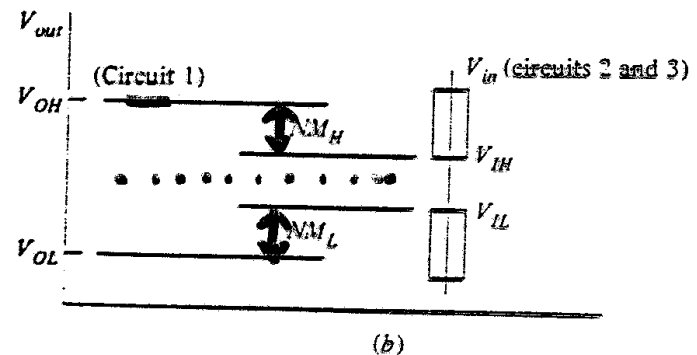
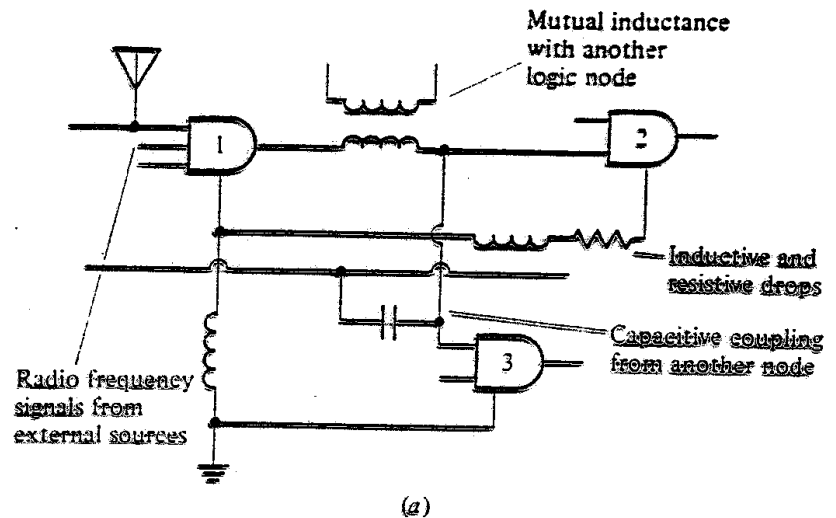


Noise Margin

- $NMH = V_{OH} - V_{IH}$
- $NML = V_{OL} - V_{IL}$
- Total Noise Margin (NM)

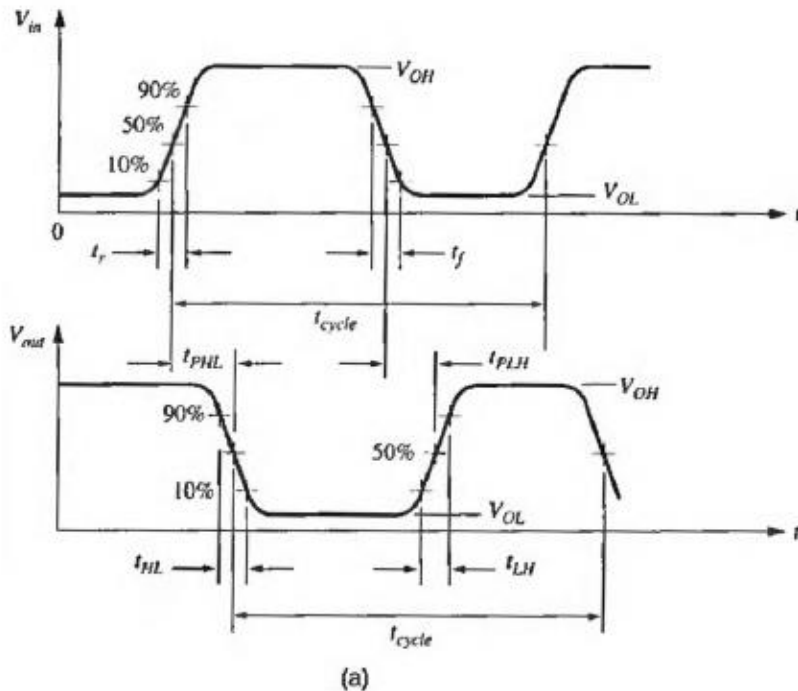
$$NM = NMH + NML$$

- Would like **signals restored** and **noise attenuated** when passing through digital circuits

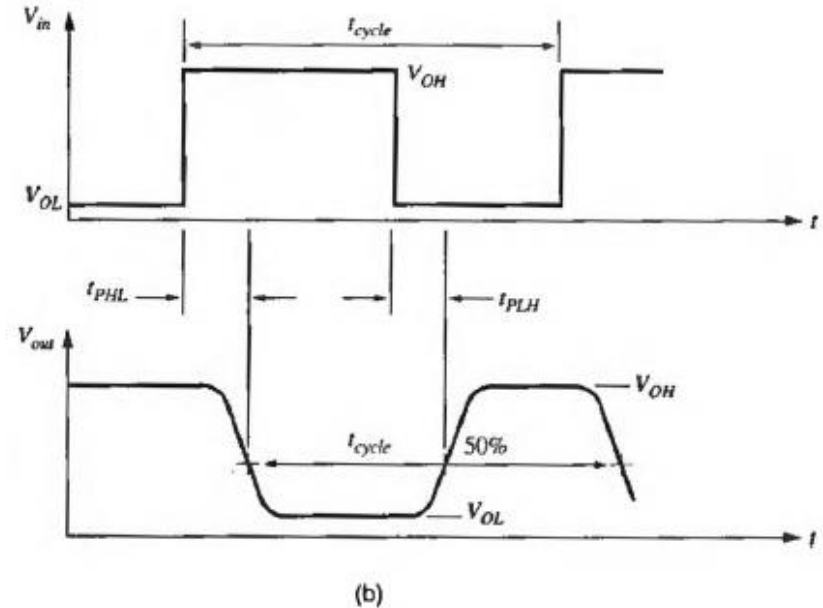


Transient Characteristics

Actual



Ideal



- t_r, t_f : 10 to 90%; $t_{PHL}, t_{PLH} \sim 50\%$; $t_{cycle} = 1/f_{clock}$
- Circuit speed determined by charging or discharging of capacitors, esp. MOSFETs

$$Q = C V; I = C \Delta V / \Delta t \text{ or } \Delta t = C \Delta V / I$$

Power Estimation

- Basic Power Equation

$$P = I \times V_{DD}$$

- Total Power

$$P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}}$$

$$P_{\text{static}} = (I_{DC} + I_{\text{leakage}}) V_{DD}$$

$$P_{\text{dynamic}} = C V_{DD}^2 f$$

- Tradeoff

Performance vs. Power

